AlHudah

Quran Platform

Complete Implementation Guide

alhudah.com | April 2026

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The Mission

Making the Holy Quran accessible, searchable, and beautifully presented through modern technology

114	6,236	30	50+
Surahs	Verses	Juz	Translations

Technology Stack

Frontend

HTML/CSS/JS

Zero build step, fast

Database

Supabase

PostgreSQL + REST API

API

quran.com API v4

50+ translations & tafsir

AI

OpenAI GPT-4o-mini

Semantic search & chat

Hosting

Cloudflare Pages

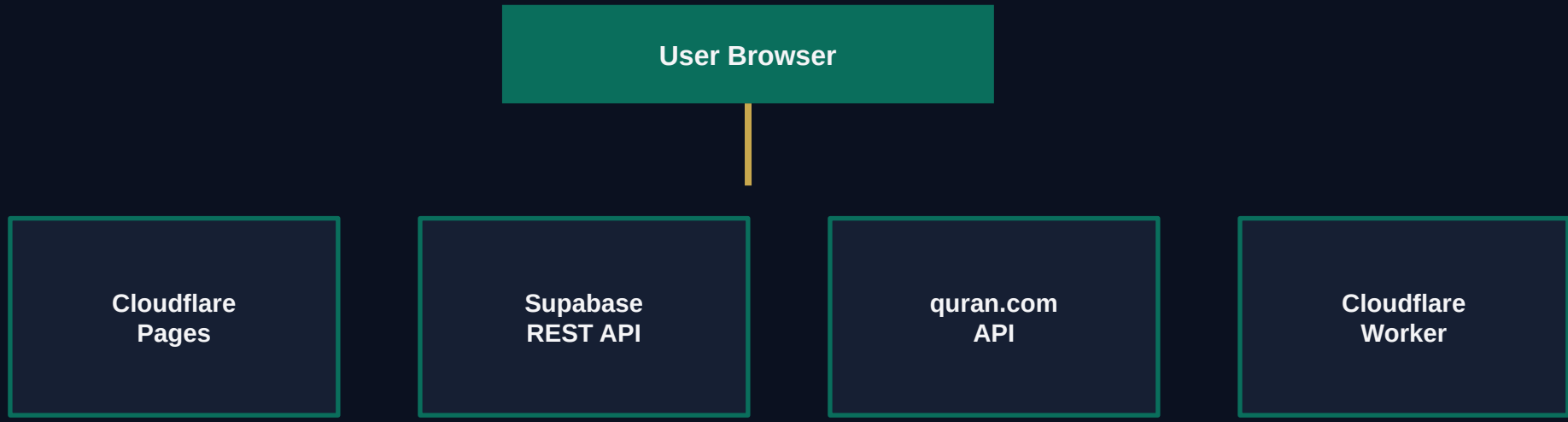
Free global CDN

Auth

Supabase Auth

Email + Google OAuth

System Architecture



APIs & Data Flow

Monthly Cost Estimate

Service	Free Tier	Estimated Cost
Cloudflare Pages	Unlimited sites	$0
Supabase	500MB, 50K req	$0
Cloudflare Worker	100K req/day	$0
OpenAI API	Pay per use	~$2
Domain (annual)	~$12/year	~$1/mo
TOTAL		$3/month

$3/mo

Infrastructure Setup

Setup

1

Register Domain

alhudah.com on Cloudflare

2

GitHub Repo

Initialize repository

3

Connect Pages

Link to Cloudflare Pages

4

SSL & DNS

Configure security headers

Project File Structure

```
alhudah/
  index.html          (Homepage)
  surah.html          (Verse reader)
  search.html         (AI search)
  about.html          (About page)
  css/
    style.css         (Main styles)
    theme.css         (Light/dark)
  js/
    app.js            (Core logic)
    api.js            (Supabase REST)
    ai-search.js      (OpenAI integration)
    service-worker.js (PWA caching)
  _headers            (CSP, cache rules)
  .github/workflows/
    validate.yml      (CI checks)
```

Policy

- default-src 'self'
- script-src 'self' cdn.jsdelivr.net
- style-src 'self' 'unsafe-inline' fonts.googleapis.com
- font-src fonts.gstatic.com
- img-src 'self' data: https:
- connect-src 'self' api.supabase.co quran.com api.openai.com

Always include fonts.googleapis.com in connect-src

Database Design

Database Schema

surahs

114

surah_number, name_ar, name_en

ayahs

6,236

ayah_number, surah_id, text_ar

translations

50+

ayah_id, language, text

tafsir

1,000+

ayah_id, scholar, content

bookmarks

Dynamic

user_id, ayah_id, created_at

Surahs & Ayahs Tables

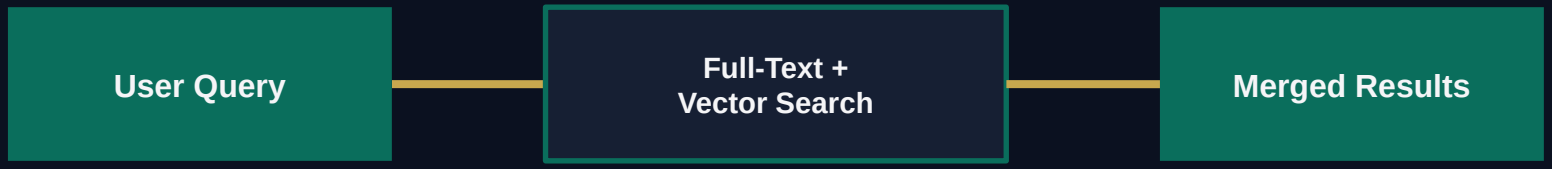
Surahs

Column	Type	Description
surah_number	INT	1-114
name_ar	TEXT	Arabic name
name_en	TEXT	English name
verses_count	INT	Verse count

Ayahs

Column	Type	Description
ayah_number	INT	1-6,236
surah_id	FK	Link to surah
text_ar	TEXT	Arabic text
juz	INT	Juz number (1-30)

Hybrid Search System



Full-Text Search:
Fast keyword matching on Arabic text and translations

Vector Search:
Semantic understanding using OpenAI embeddings

Combined: Best of both — speed + semantic meaning

(RLS)

Table	Operation	Policy
bookmarks	SELECT	auth.uid() = user_id
bookmarks	INSERT	auth.uid() = user_id
bookmarks	DELETE	auth.uid() = user_id
ayahs	SELECT	Public (anon key OK)
translations	SELECT	Public (anon key OK)
tafsir	SELECT	Public (anon key OK)

RLS ensures user bookmarks are private while Quran data is public and fast.

Data Population

Quran Data Sources

Tanzil.net

Arabic text:
XML format with full diacritics

Translations:
30+ languages, offline-friendly

Format:
Download & parse locally

Speed:
Instant (no API call)

quran.com API v4

Translations:
50+ with scholar info

Tafsir:
6 major scholars available

Format:
JSON REST API, live updates

Speed:
Requires API call (~200ms)

Endpoints

Endpoint	Purpose
/chapters	Get all 114 surahs with metadata
/verses/by_chapter/{id}	Verses for a surah (paginated)
/chapters/{id}/info	Surah introduction & details
/translations?language=en	List available translations
/tafsirs	List tafsir sources (Ibn Kathir, etc)
/ayahs/{id}/tafsirs?tafsir_id=13	Tafsir for a specific verse

Data Pipeline

Source
Tanzil XML / quran.com API

Parse
Extract structured data

Transform
Map to schema

Insert
Bulk load to Supabase

Verify
Validate row counts

Frontend

Design System

Teal

Gold

Dark

Light

Typography

Titles:
Arial Black, 36-44pt
Body:
Calibri, 14-16pt
Code: Consolas, 12pt

Spacing

Margins: 0.5in minimum
Gaps: 0.3-0.5in between blocks
Line height: 1.5
Card padding: 0.2in

Site Pages

Home

Landing page

Surah

Verse reader

Search

AI search

Bookmarks

Saved verses

About

Info page

Privacy

Policy

Terms

Terms of use

404

Not found

Offline

PWA cache

Homepage Layout

Hero	Large Quran heading + tagline
Search	Full-width search bar + AI power
Featured	Verse of the day
Grid	Browse 114 surahs

Light & Dark Theme

Light Mode

Dark Mode

Progressive Web App

1

Installable

Add to home screen, desktop app experience

2

Offline Mode

Service Worker caches core assets & data

3

Push Notifications

Notify users of new tafsir or features

Quran Reader

Browse 114 Surahs

1

الفاتحة

Al-Fatihah

7 verses

2

البقرة

Al-Baqarah

286 verses

3

آل عمران

Ali 'Imran

200 verses

... and 111 more surahs with instant navigation

Reading

Al-Fatihah 1:1

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

In the name of Allah, the Most Gracious, the Most Merciful

Urdu: اللہ کے نام سے جو بہت مہربان اور بہت رحم کرنے والا

Bookmark | Tafsir | Share | Copy

Reading Navigation

Juz Selector

Jump between 30 sections

Prev/Next Surah

Chapter-by-chapter browsing

Jump to Verse

Go to any ayah instantly

Progress Bar

Track reading position

Save & Share

Bookmarks

- One-click save
- Synced to account
- View all anytime
- Export as notes

Sharing

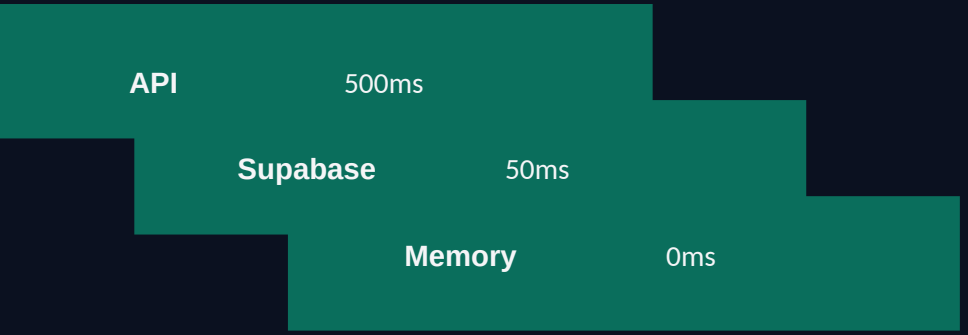
- Web Share API
- Social media cards
- Verse link with meta
- Direct copy to clipboard

Tafsir System

Multiple Tafsir Sources

Scholar	Language	Description
Ibn Kathir	Arabic/English	Classical, comprehensive
Jalalayn	Arabic/English	Concise & detailed
Maududi	English/Urdu	Modern interpretation
Saadi	Arabic/English	Linguistic analysis
Al-Tabari	Arabic	Historical context
Qurtubi	Arabic	Legal rulings

3-Level Cache



Smart caching ensures Quran data loads instantly to users worldwide

AI Search

Search

Question
User natural language query

Embedding
Convert to vector (OpenAI)

Search
Full-text + semantic match

Rank
Merge & score results

Answer
GPT-4o-mini response

Example: 'What does Islam say about mercy?' → Searches ayahs with related concepts → Returns verses with AI explanation

Architecture

API Key Protected:
OpenAI key stored in Cloudflare Worker secret, never exposed to browser

Rate Limiting:
Prevent abuse and excessive API costs

Request Validation:
Sanitize inputs before forwarding to OpenAI

Voice Search

Technology:
Web Speech API (native browser)

Supported Browsers:
Chrome, Edge, Safari on iOS 14.5+

Accuracy:
95%+ for English & Arabic

Fallback:
Text input if microphone unavailable

Search Cost Analysis

Operation	Cost per 1000	Monthly (10K searches)
Embedding (1,536 dims)	$0.002	$0.02
Chat Completion	$0.15	$1.50
TOTAL		~$1.50/mo

~$0.02 per 1000 searches

SEO & Performance

Optimization

Meta Tags

Title, description, OG tags per page

JSON-LD Schema

WebSite, Book, FAQPage markup

Semantic HTML

Proper heading hierarchy & structure

Sitemap

Auto-generated, submitted to GSC

Performance Targets

LCP	CLS	INP
< 2.5s	< 0.1	< 200ms
Largest Contentful Paint	Cumulative Layout Shift	Interaction to Next Paint

Strategy:
Image optimization, lazy loading, service worker caching, CSS/JS minification, edge caching via Cloudflare

Deployment

CI/CD Pipeline

Git Push
Code committed to main

Validate
GitHub Actions runs tests

Deploy
Cloudflare Pages builds

Live
Global CDN serves site

Zero-downtime deployment: Changes push to main → auto-deploy to Cloudflare Pages in ~30 seconds

Launch Checklist

1. Domain live with SSL certificate configured
2. Cloudflare Pages deployed and tested
3. Supabase project created with full schema
4. OpenAI API key configured in Worker secret
5. Service worker caching working offline
6. All metadata & SEO tags implemented
7. Mobile responsiveness verified (all screen sizes)
8. Lighthouse score 90+ on all metrics
9. Analytics & error tracking enabled
10. Backup & monitoring configured

Timeline & Summary

Phased Execution

Phase 0-1	Setup & Database	Week 1
Phase 2-3	Frontend & Reader	Week 2
Phase 4-5	Tafsir & Search	Week 3
Phase 6-7	SEO & Deploy	Week 4
Phase 8-9	QA & Launch	Week 5

Hadith

CSP Headers

Always include font domains in connect-src for Google Fonts

Theme Flash

Apply theme before first paint to prevent flicker on page load

Service Worker

Test font caching separately; font failures break offline

Arabic Fonts

Cairo Play needs explicit font-weight declaration for bold

Git Workflow

114 Surahs | 6,236 Verses | 30 Juz

50+ Translations | 6 Tafsir Sources

Hybrid AI Search | ~$3/month

Thank You

alhudah.com

GitHub: ehsaasradio-bot
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